

DSA Exam Notes

Recursion & Backtracking

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1. Recursion kya hoti hai?

- Recursion tab hoti hai jab ek function apne aap ko call karta hai.
- Iska use problem ko chhote parts mein solve karne ke liye hota hai.
- Har recursive call ke liye ek base case hona chahiye.

2. Important terms

- Base Case: Jab recursion stop ho jaye.
- Recursive Case: Problem ko chhota karke phir se call karna.
- Call Stack: Function calls ka memory stack.

3. Recursion ke rules

1. Base case likho.
2. Problem ko smaller version mein todho.
3. Recursive call ko base case ki taraf le jao.
4. Overlapping calls se bachao.

4. Common examples

- Factorial
- Fibonacci
- Sum of digits
- Power calculation
- Binary search (recursive)

5. Time and Space Complexity

- Factorial: $O(n)$ time, $O(n)$ stack
- Fibonacci: $O(2^n)$ naive, $O(n)$ optimized DP
- Binary search: $O(\log n)$

6. Types of recursion

- Direct recursion
- Indirect recursion
- Tail recursion
- Head recursion

7. Backtracking kya hoti hai?

- Solution ko step by step build karte hain.
- Agar current choice galat hoti hai to wapas jaate hain.
- Isse searching aur combination problems mein use hota hai.

8. Backtracking ke important points

- Choice choose karo
- Constraint check karo
- Agar valid hai to recurse karo
- Agar fail ho to undo karo

9. Famous problems

- Permutations
- Subsets
- N-Queens
- Rat in a Maze
- Sudoku

10. Example: Factorial

```
```cpp
int fact(int n) {
 if (n == 0) return 1;
 return n * fact(n - 1);
}
```
```

11. Example: Permutations

- Ek element choose karo

- Baki elements ke saath recursion karo
- Backtrack karke next choice lo

12. Exam tips

- Base case hamesha dhyan mein rakho
- Stack overflow ka dhyan rakho
- Time complexity samjho
- Dry run karna seekho
- Backtracking mein undo/restore zaroori hota hai

Prepared for exam revision